P510/1
PHYSICS
Paper 1
July/Aug. 2026
3 hours

GLORISO EXAMINATIONS BOARD (GEB)-KAMPALA

SECONDARY SCHOOLS JOINT MOCK ASSESSMENTS, 2026

Uganda Advanced Certificate of Education

PHYSICS

Paper 1

3 hours

INSTRUCTIONS TO CANDIDATES:

- ✓ *This examination paper consists of two sections;* **A** *and* **B**. *It has* **six** *items.*
- ✓ *Section* **A** *has* **two compulsory** *items.*
- ✓ *Sections* **B** *has two parts;* **I** *and* **II**. *Respond to only* **one** *item from each part.*
- ✓ *Respond to* **four** *items in all.*
- ✓ *Any additional item (s) responded to will* **not** *be scored.*
- ✓ *Responses to* **each** *item should start on a* **fresh page**.
- ✓ **All** *responses must be written in the answer booklet (s) provided.*
- ✓ *Silent Non-programmable scientific calculators may be used.*

SECTION A.

This section has **two compulsory** *items.*

Item 1.

A regional referral hospital recently installed a digital X-ray machine for diagnosing bone fractures and chest infections. During commissioning, medical physicists conducted several tests to ensure that the X-rays produced were of the required wavelength and intensity. The X-ray tube operated at a potential difference of **80 kV** and a tube current of 15 mA. A crystal spectrometer with spacing **2.8×10^{-10} m** was used to analyze the X-rays. A first-order diffraction maximum was observed at an angle of **13.5°.** The detector system is periodically calibrated using a radioactive source with an initial activity of **4.8×10^{8}** disintegration per second. The source has a half-life of 6 years and has been in use for **18 years**. The hospital administration wishes to determine whether the equipment is operating effectively and to ensure the safe application of radiation in diagnosis. 2

Task:

a) Describe, with the aid of a labelled diagram, the structure and operation of an X-ray tube. Hence explain the production of X-rays and state four medical applications of X-rays.

b) Using the information provided,

(i) determine the wavelength of the X-rays produced;

(ii) Determine the maximum photon energy and the minimum wavelength of the X-rays and hence comment on the suitability of the X-rays for diagnostic medical imaging.

(c) (i) Determine the present activity of the radioactive calibration source and Hence comment on its suitability for detector calibration.

(iii) Explain the importance of radioactive decay and radiation safety in medical diagnosis.

Item 2.

A construction company in Kampala uses a steel concrete pan attached to a motorized winch and pulley system to transport concrete and other building materials to the upper floors of a multi-storey building. During a performance test, the loaded pan together with its contents had a total mass of **250 kg.** Starting from rest, the pan was raised vertically through a height of **18 m** in **6.0 s** with uniform acceleration.

Engineers estimated that the total resistive force due to friction in the pulley bearings and air resistance was **180 N.** The steel cable supporting the pan had an original length of **12 m** and a cross-sectional area of **3.2×10^{-5} m².**

During operation, the cable extended by **5.4×10^{-3} m.** The pulley support was fixed to a frame held in equilibrium by two steel stays inclined at **40°** and **55°** to the horizontal. The company management wished to determine whether the lifting system was operating efficiently and whether the cable and supporting frame were suitable for continuous use. They also wanted to verify the validity of some physical relationships used by the engineers in designing the system.

Task:

a) (i) Using the information provided, analyse the motion of the loaded pan and determine the power developed by the winch.

(ii) Hence explain how friction and gravity influence the efficiency and safe operation of the lifting system.

b) Determine the mechanical properties of the supporting cable and the forces in the supporting stays. Hence assess the suitability of the cable and the safety of the supporting frame for prolonged use.
c) Verify that the relation **P = Fv** is dimensionally consistent and explain the importance of dimensional analysis in engineering design and construction.

SECTION B
Part I
*Respond to **only one** item from this part.*

Item 3.
A media and entertainment company wishes to establish a modern recording studio for producing radio programmes, podcasts and live musical performances. To improve visibility and communication between the control room and the recording room, large plane mirrors and transparent glass partitions were installed. Video cameras fitted with converging lenses were also mounted to monitor performers during recording sessions.
The studio uses loudspeakers whose diaphragms vibrate continuously to produce sound. During a sound check, it's observed that some musical notes became unusually loud and persistent, affecting the quality of recordings. Measurements showed that one of the loudspeakers produced 200 vibrations per second and that the speed of sound in air within the studio was estimated to be **340 m s^{-1}.**
The designers wish to understand how the behavior of light and sound could be utilized to improve image formation, communication and acoustic quality in the studio.
Task:
a) Explain how reflection and refraction of light together with the operation of a camera contribute to effective monitoring and communication in the studio.
b) Determine the period and wavelength of the sound produced by the loudspeaker and hence explain how simple harmonic motion, progressive waves and stationary waves influence resonance and the acoustic quality of the studio.

Item 4.
A dairy processing company produces pasteurized milk and yoghurt for distribution to different parts of the country. To maintain product quality, freshly collected milk is heated and cooled under controlled conditions. Digital thermometers and temperature sensors are used to monitor the temperature during processing.
During one production cycle, **80 kg** of milk initially at **18°C** was heated to **72°C** before being transferred into insulated storage tanks. Steam from a boiler supplies heat to the milk. Technicians observed that some heat is lost to the surroundings despite the use of insulation. They also noted that changes in temperature affect the pressure of the steam and consequently the performance of the heating system.
[Take the specific heat capacity of milk to be 3.9 × 10^3 J kg^{-1} K^{-1}.]
Task:
a) Determine the quantity of heat required to raise the temperature of the milk and hence explain how heat is transferred within the system and how heat losses may be minimized.
b) Explain how temperature changes affect the pressure and volume of the steam in the

boiler and hence comment on the importance of controlling temperature and pressure during milk processing.

Part II

*Respond to **only one** item from this part*

Item 5.

A solar-powered cold room used for preserving fruits and vegetables employs an inverter system to convert direct current from batteries into alternating current required by compressors, lighting units and control devices. To maintain a steady output voltage, the inverter contains capacitor banks which store electrical energy and smooth fluctuations in the supply. During periods of low humidity, maintenance personnel occasionally observe sparks on insulated surfaces due to the accumulation of static charges. The inverter delivers a $240\ V, 50\ Hz$ supply and uses a capacitor bank of capacitance **2500 μF.**

4

Task:

a) Explain the production of electrostatic charges and the principle of operation of a capacitor. Hence establish the relation $Q = CV$ and determine the charge stored by the capacitor bank.

b) Describe the nature of alternating current and hence establish the relation $T = \frac{1}{f}$ Comment on the importance of capacitors and alternating current in modern electrical systems.

Item 6.

A commercial greenhouse uses an automated irrigation system in which moisture sensors control electromagnetic relays that switch water pumps on and off. Each relay consists of a coil of **800 turns** wound uniformly around a **soft iron core** of mean length **0.20 m**, cross-sectional area **1.0×10^{-4} m²**, and relative permeability **150.**The relay is connected to a **24 V d.c. supply,** and the resistance of the coil is **12 Ω**.

During operation, the current in the coil falls uniformly from **2.0 A** to zero in **4.0 ms** when the circuit is opened. A capacitor of capacitance **10 μF** is connected across the relay contacts to reduce sparking.

The digital control unit receives binary signals from two moisture sensors **A** and **B** and is programmed to switch on the pump only when both sensors indicate dry soil.

[Take $\mu_0 = 4\pi \times 10-7$ H m^{-1}]

Task:

a) Describe how magnetism is produced in the relay coil and account for the suitability of soft iron as the core material. Hence determine the current in the coil and the magnetic flux density developed in the core.

b) Explain how electromagnetic induction occurs when the current changes and hence determine the self-inductance of the coil, the induced e.m.f. and the energy stored in the capacitor. Explain how the capacitor reduces sparking.

c) Distinguish between analogue and digital signals and comment on the role of digital electronics in automated irrigation systems. Hence explain the logic condition required for the pump to switch on.

"GOOD LUCK"

****THE END****

© 2026 GLORISO-Kampala
P.O. Box 23799, Tel. 0392553142 / 0200944391, Kampala (U).

END.